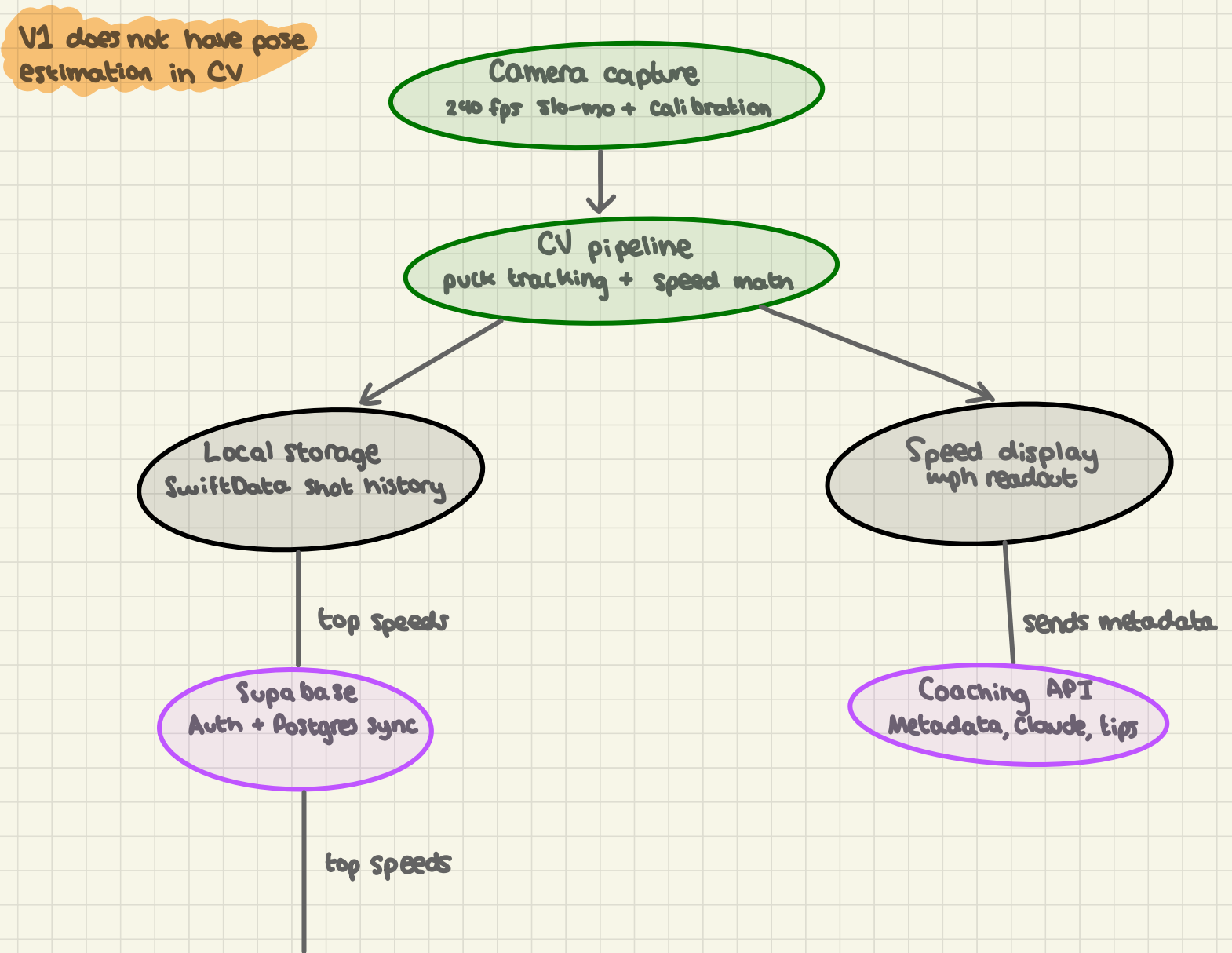

V1 does not have pose estimation in CV
Camera capture
240 fps slo-mo + calibration
CV pipeline
puck tracking + speed math
Local storage
SwiftData shot history
Speed display
mph readout
top speeds
sends metadata
Supabase
Auth + Postgres sync
Coaching API
Metadata, Claude, tips
top speeds

Leaderboards
THSHL, team, friends

Camera capture

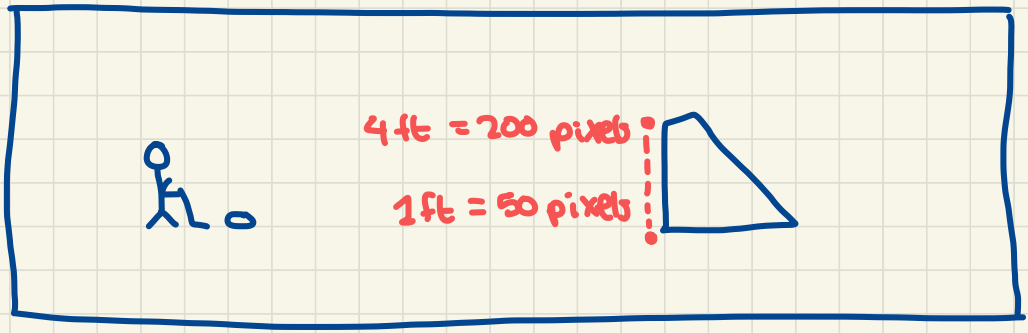
- User uses in-app camera or uploads their own video
- App automatically converts to 240fps. More frames = more samples

=> Calibration

→ User drops a dot on top of the post, a dot at the bottom

→ Manually input distance between dots

↝ What this does is that it now knows the correlation between pixels and real-life feet. If 4 feet is 100 pixels, it now knows 1 feet 25 pixels.



CV pipeline

=> Puck tracking

- → A model that looks at each frame and answers one question: "where's the puck?" (all happens internally)
- → Output is a pixel coordinate per frame
  - ↳ Frame 1: (400, 300)
  - ↳ Frame 2: (473, 296)

=> Speed math

1.) Take two consecutive frames: Puck moved 24 pixels

2.) Convert to feet using calibration. Calibration says 50 pixels = 1 foot.
So $24 \div 50$ px/ft = 0.48 ft

3.) 0.48 ft happened in 1/240th of a second. So $0.48 \times 240 = 115$ ft/second

4.) $115 \div 1.467 = 78$ mph

Local storage

=> SwiftData shot history

Keeps track of:

- Speed (the mph)
- Timestamp
- Session it belongs to

Speed display

=> mph readout

- After recording is over, it tells you the speed of every shot
- Every mph has a short shot clip attached to it to correlate the two

Supabase

=> Auth

- "Sign in with Apple"
- No need to build any password/security mess, Supabase will handle it

=> Postgres

Shot happens → Saved locally in milliseconds, history updates instantly → then in the background, whenever there is internet, the app quietly unsynced shots into supabase.

Coaching API

=> Metadata

- Gives metadata for the model to interpret

```
{
    "speed_mph": 61
    "shot_type": "wrister"
    "session_shot_number": 14
    "session_avg": 58
    "trend": "speed dropping over last 5 shots"
}
```

=> Claude/tips

- Metadata transfers to Claude which gives tips